Exemplars Move Retrieval, Instructions Move Rejection: An Anatomy of Inference-Time Interventions for Closed-Book Knowledge Base Construction

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Abstract

We describe our system for the AKBC 2026 shared task on closed-book knowledge base construction. The final system samples 20 candidates per query from Mistral Small 3.2 (24B) under two deliberately different prompts, aggregates them with relation-specific policies, and resolves disagreements between the two with one vote from an unrelated-family 8B model (Llama-3.1-8B), totalling 31.8B parameters under the task's 32B cap. It scores Macro F1 0.641 on validation and 0.625 on the official test set, against 0.313 for the baseline. The larger part of the paper is a mechanism study over two base models and roughly twenty-five controlled experiments, asking which inference-time interventions move a closed-book system and which only appear to. The findings: (1) how reliably a model recalls the value stored for an entity label dominates every intervention we tried—a 12-row probe predicts system quality across ten models, and switching models moved the score more than nine intervention experiments combined; (2) recall inside a candidate is a single early anchor-sampling event whose distribution is moved by *exemplar style*, not instruction text: two in-prompt exemplars demonstrating a one-sentence verbalized recall shift the answer mode by +6 rows, while the same request as an instruction, under bare-JSON few-shot exemplars, moves nothing; (3) this control is conditional—null where recall is unreliable, harmful on an unreliable relation, severed by RL post-training; (4) selection within one model's candidate set has no usable signal, but arbitrating between the two prompts with an external-family 8B model gains rows on validation *and* test, where an isolated ablation confirms the transfer; (5) validation gains split cleanly into those that transfer (mechanism-level) and those fitted to validation row statistics, and a 100-row validation set cannot tell them apart.

1 Introduction

The LM-KBC shared task series (LM-KBC organizers, 2023, 2024) probes how much factual knowledge language models can turn into knowledge base entries: given a subject and a relation, a system must output *all* correct object values—possibly none, one, or hundreds—under closed-book conditions: no retrieval, no external corpora, no fine-tuning, and at most 32B neural parameters at inference.

Under these conditions the only controls are at inference time: model choice, prompts, sampling, and aggregation over samples. We entered the task twice, in effect: first with a Qwen3.5-27B thinking system (test 0.571), then—after a systematic model screen—with a Mistral Small 3.2 system whose final form adds a second prompt and an 8B cross-family arbiter (test 0.610). Alongside the system we report the controlled experiments behind each decision, including seven submissions whose test scores give us what validation cannot: isolated ablations and transfer measurements on the hidden set. The picture that emerges:

1. **Reliable label-to-value recall dominates.** A 12-row probe (30 GPU-minutes per model) predicts downstream quality across ten models; switching from Qwen to Mistral beat all our prompt, sampling, and selection experiments combined (Sec. 4.1).

2. **Recall is a single early anchor-sampling event** (Sec. 4.2), replicated across families.

3. **Exemplar style moves the anchor distribution; instruction text does not** (Sec. 4.3)—conditional on recall reliability and on post-training (Sec. 4.4).

4. **Within one model, selection has no usable signal; across prompts and families it does** (Sec. 4.5).

5. **Validation gains split into transferring and non-transferring types**, by mechanism, not effect size (Sec. 4.6).

6. **The residual errors are wrong values inherited from the training corpus**: stable within a model, changing across generations, shared across families, and *amplified* by instruct-tuning (Sec. 4.7).

2 Task and Constraints

The 2026 task covers six relations over 478 validation and 477 test subject–relation pairs: two set-valued relations that may be empty (`countryLandBordersCountry`, `companyTradesAtStockExchange`), one single-valued string relation that may be empty (`personHasCityOfDeath`), one large-set relation (`awardWonBy`), and two numeric relations (`hasArea`, `hasCapacity`) scored at 5% relative tolerance. Gold values come from Wikidata (areas `P2046`, capacities `P1083`), with alias lists accepted for strings. The official "All Relations" score averages over *rows*, not relations, so the four 100-row relations dominate. Systems are closed-book and capped at 32B inference parameters, quantization notwithstanding.

3 System Description

3.1 Generation

The final system generates with Mistral Small 3.2 24B (BF16 GGUF, a lossless repackaging of the released weights), served by `llama.cpp` with pinned revisions, 20 candidates per query, temperature 0.6, top-p 0.95, deterministic per-candidate seeds, and no extended thinking (Sec. 4.4 explains why thinking is null here). Numeric relations use *two* prompts of deliberately different style:

Recall prompt (`hasArea`): no few-shot examples; instead the relation instruction carries two in-prompt exemplars demonstrating a labeled recall format (`Entity:` one sentence identifying the referent; `Memory:` "I recall that the Wikipedia infobox lists X's total area as Y"; `Unit:` a conversion check; `Answer:` a bare JSON array). Section 4.3 shows the exemplars, not the instructions, are the active ingredient.

Grounding prompt (`hasArea`, `hasCapacity`): five standard few-shot examples plus an instruction naming the recall target ("the value stored in Wikidata property `P2046`/`P1083`, as shown in the Wikipedia infobox, for the entity with exactly this name"), with digit-preservation and disambiguation clauses for capacities. String relations use the plain task prompt with five few-shot examples. `hasCapacity` adds a third prompt at zero parameter cost: the *same* 24B model queried as raw text completion (an infobox-style prefix continued without the chat template), which surfaces exact stored figures that chat prompting does not (Sec. 4.7).

3.2 Aggregation and Arbitration

Numeric candidates are clustered at 5% relative tolerance; `dominant_cluster` predicts the median of the best-supported cluster, and `unit_equivalence` first merges clusters related by unit-conversion factors. `personHasCityOfDeath` uses majority voting with a minimum-vote abstention gate; `companyTradesAtStockExchange` uses a frequency threshold (0.5) with empty-majority abstention, after merging surface variants—case variants and acronyms that spell a co-occurring long form (NYSE $\rightarrow$ New York Stock Exchange)—into a single count, repairing vote splits that would drop majority answers below threshold. `awardWonBy` keeps a recall-oriented threshold plus a unanimous-no verification filter from the Qwen-era system.

Cross-family arbitration. When the two prompts' aggregated answers for a numeric row disagree by more than 5%, the system asks a third, unrelated-family model—Llama-3.1-8B, whose own answers are far weaker (Table 2)—for its modal value over 20 samples, and switches to the secondary answer only if the arbiter matches it and not the primary one. The design cannot lose rows by construction: both options are strong-model answers, and no match keeps the default. Total parameters: $24\text{B} + 8\text{B} \approx 31.8\text{B} \leq 32\text{B}$.

Tolerance-geometry bridge. Finally, each `hasArea` answer v is shifted within its own $\pm 5\%$ window to the point covered by the most candidate-interval vote mass across both prompts. Two values whose ratio is below $1.05/0.95 \approx 1.105$ admit a single prediction correct for either, so the bridge covers a runner-up cluster without giving up the mode, with no signal identifying which minority value is correct (Sec. 4.5). It follows from the metric geometry, has no tuned constants, and gains $+1$ to $+2$ rows across every prompt pairing and arbiter seed we tested (82/81/83/82 from a base of 80).

System	val	test
Official baseline (Qwen3.5-9B)	0.313	—
Qwen3.5-27B thinking system	0.588	0.571
Mistral 3.2, grounding only (base)	0.595	0.595
v3: + recall prompt, area arbitration, company surface merge @0.5	0.629	0.604
v4: + capacity arbitration	0.633	0.606
v5: + acronym merging	0.635	0.610
v6: + second capacity arbiter†	0.637	0.610
v7: + city merge (rejected)	0.660	0.608
v8: dataset-release re-keying*	0.637	**0.625**
v9–v10: + area bridge; compliant completion prompt replaces †	0.641	pending

Table 1: System progression (Macro F1). Each vn differs from its predecessor in a small, known row set, so the test column doubles as a sequence of isolated ablations on the hidden set (Sec. 4.6). *The August subject-disambiguation release renamed 15 test subjects and dropped two rows (477→475); submissions up to v7 score the renamed rows as zero, understating them by $\approx$ 0.015. †A second 8B arbiter that put v6–v9 over the 32B cap; v10 replaces it with the zero-parameter completion prompt at equal validation score (see Limitations).

3.3 Results

Table 1 traces the system. The Qwen→Mistral switch is worth $+0.024$ on test with the rest of the pipeline unchanged; the Mistral-era interventions and the dataset re-keying add $+0.030$ more, of which every transferring component is mechanism-level (Sec. 4.6). Per-relation test scores at v8: borders 0.958, area 0.800, company 0.778, city 0.510, award 0.279, capacity 0.214.

4 Anatomy of Inference-Time Interventions

All experiments change one stated condition; numeric testbeds are `hasArea` and `hasCapacity` (100 validation rows $\times$ 20 candidates per prompt). **Oracle** is the number of rows whose candidate set contains at least one within-tolerance value; the **mode** row count is what `dominant_cluster` selects. The **anchor** is the first concrete value for the queried quantity in a candidate.

4.1 Reliable Label-to-Value Recall Dominates; a 12-Row Probe Measures It

Our Qwen-era system plateaued after nine intervention experiments (prompts, temperature, output routes, thinking budgets, judge models, log-probability confidence) whose combined deployable gain was under $+0.01$. A 12-row probe of that system's wrong rows, run across candidate models (Table 2), showed Mistral Small 3.2 answering 11 of the 12 somewhere in its candidate set. Re-generating the pipeline on Mistral with nothing else changed moved validation 0.588→ 0.595 and test 0.571→0.595—more than everything else we had tried combined. This is the task's first-order fact: closed-book KBC quality is set by which label-to-value pairings pretraining left intact, and a half-hour probe measures it before any tuning.

Model (12-row probe)	any	mode
Mistral Small 3.2 Instruct	**11**	**8**
Mistral Small 3.1 Base	10	7
Mistral Small 3.1 Instruct	9	4
Llama-3.1-8B Instruct	7	4
Aya Expanse 32B	3	1
Qwen3-8B	1	0

Table 2: The recall probe: 12 `hasArea` rows stratifying the Qwen-era system's failures, 20 candidates each; *any* = rows with a correct candidate, *mode* = rows whose modal cluster is correct. Qwen3.5/3.6-27B, Gemma 4 31B, GLM-Z1-32B and llm-jp-4-32b (not shown) sit between Llama-8B and Aya. The probe costs $\sim$30 GPU-minutes per model and its ranking matched full-validation quality wherever we deployed.

4.2 Recall Is a Single Early Anchor-Sampling Event

In persisted reasoning traces of the Qwen-era system, the answer value first appears at a median 17% of the trace, in 95–98% of candidates inside a retrieval template ("Commonly cited: X"); the remaining 80% of the trace never dislodges it. The verification text is conditioned on the anchor rather than checking it: fabricated Wikidata Q-IDs and cross-language "confirmations" agree with whatever the anchor was, and wrong candidates carry *more* of this apparatus than correct ones (6.9 vs. 4.2 source mentions). The anchor mechanism replicates on Gemma 4 31B and Qwen3.6; the confabulated verification does not (it is Qwen's trained style). Extended thinking is thus an *exploration* aid: it raises the oracle where recall is unreliable (Qwen $+9$ rows, Gemma $+7$) and is null on Mistral and its RL-tuned sibling Magistral.

4.3 Exemplar Style Moves Retrieval; Instruction Text Does Not

The single largest prompt effect we found—`hasArea` mode $72 \rightarrow 78$, transferring to test as $0.70 \rightarrow 0.75$—looked at first like an instruction effect: a step-structured expert prompt, adapted

`hasArea` prompt (Mistral 3.2)	Few-shot examples bare-JSON $\times 5$	none
grounding instruction	71 (o. 89)	**41** (o. 59)
recall-format text	72 (o. 83)	**78** (o. 84–89)

Table 3: The $2{\times}2$ comparison: mode-correct rows out of 100 (oracle in parentheses). The recall-format text carries two exemplars demonstrating the recall step inside the instruction, so the bottom-right cell keeps two exemplars while the top-right has *none* and collapses to 41. At fixed example style the instruction text moves the result by ± 1; the example style moves it by $+6$ / -30.

Component (mechanism)	val	test	transfers
recall exemplars (§4.3)	$+6$	$+5$	✓
company surface merge (§3.2)	$+1$	$+2$	✓
8B arbitration (§4.5)	$+2$	$+1$	✓
capacity prompt (best of 6)	$+2$	-2	×
second arbiter (increment)	$+1$	0	×
city two-stage merge	$+11$	-1	×

Table 4: Row-level validation and test deltas for every deployed component, from the isolated ablations of Table 1. The split follows the mechanism, not the effect size: our largest validation gain ($+11$ rows) transferred worst.

from a style published by another participant and re-derived in our own wording, beat our grounding instruction by 6 rows. The $2{\times}2$ comparison (Table 3) shows the instruction text is nearly inert. What matters is the *exemplars*: with two in-prompt exemplars demonstrating a one-sentence verbalized recall ("I recall that the Wikipedia infobox lists Luxembourg's total area as 2586.4 km^2"), 99% of candidates reproduce the verbalization and the mode moves $+6$; with five bare-JSON few-shot exemplars, 0% verbalize—the same procedure requested as instruction text is silently ignored—and the mode does not move; with no examples, output collapses (41). Iterating the wording (persona, fixed phrase, provenance vocabulary) changed nothing; reproducing the labeled structure with our own labels recovered the full effect (78, at the highest oracle, 89). Examples set not just the output format but the focus of the recall distribution itself, and the gain is seed-stable (78 ± 1, single-row flips).

4.4 The Effect Is Conditional: Recall Reliability and Post-Training

Verbalized recall is a *focusing* intervention—it sharpens the anchor distribution around the most frequently recalled value (oracle drops $89 \to 84$ as the mode improves). Its sign therefore follows how reliable the model's recall already is, in mirror image to thinking: thinking explores, gaining $+9$ rows where recall is unreliable (Qwen) and nothing where it is reliable (Mistral); verbalized recall focuses and does the opposite (zero or negative on Qwen and on weak relations, $+6$ on Mistral).

On Qwen3.5 the same demonstrating exemplars take effect (96% verbalization) but move nothing; on `hasCapacity`, Mistral's weakest relation, they *reduce* oracle $60 \to 47$. Post-training removes the effect entirely: Magistral (Mistral's RL-tuned reasoner) copies the verbalized form at 99.9% compliance—in its thinking channel, in a scripted thinking channel, and in the answer body—yet its anchor distribution does not move in any of the three (70–72 throughout). The dominance of exemplars is a property of SFT-instruct models; RL fixes the response policy and disconnects style from retrieval.

4.5 Selection Has No Usable Signal Within a Model; Cross-Family Arbitration Does

Within one model's candidate set, minority correct values are indistinguishable from minority wrong ones on every feature we measured (votes, dispersion, trace features, log-probabilities; base rate 8%), which is why judges, confidence weighting, and consensus schemes all failed. What finally extracted signal is *disagreement between prompts of different style*: on the $\sim$25 rows where the recall and grounding prompts disagree by more than 5%, one vote from an unrelated-family 8B model decides for the correct side at 3:1. On validation this yields `hasArea` $78 \to 80$ $(+3/-1)$, reproduced across three prompt pairings and two arbiter seeds, and `hasCapacity` $26 \to 28$ $(+2/-0)$. Controls locate the conditions: same-style pairs produce too little disagreement to arbitrate; promoting minority clusters by the arbiter's mode is *negative*—an 8B cannot overrule a 24B mode, only break ties; and a weak arbiter (Ministral-8B, same family, mode accuracy 23 vs. Llama's 53 of 100) gains nothing. On test, the isolated v3→v4 ablation—four capacity rows changed, nothing else—confirms the transfer: capacity F1 $0.204 \to 0.214$.

4.6 Which Validation Gains Transfer

Seven submissions give per-component test measurements (Table 4). The transferring components share one property: their gain has a causal account that does not reference validation row statistics—an exemplar style that shifts the anchor distribution globally, a repair of a counting artifact (vote splits across surface forms that gold aliases accept either way), an arbitration that cannot lose rows. The non-transferring ones are all fits to validation row distributions: the best of six capacity configurations (winner's curse), an increment to an already-saturated arbiter, and—most instructive—a city pipeline whose $+11$-row gain rested on the rate at which split-vote rows happen to be correct-mode, a distributional property that differed on test (-1 row); a broad threshold plateau did not protect it. A 100-row validation set cannot detect this class of overfitting; the mechanism account can, in advance. Late in the effort we probed Wikidata-provenance instructions for the two string relations: the company variant was net-negative on validation, and the city variant showed the failed merge's profile (a narrow abstention peak, $+9/-4$ flips), which the taxonomy predicts would not transfer—we left it undeployed.

4.7 The Residual Errors Are Wrong Values Learned From the Corpus

What remains is not reasoning failure. Corfu (gold 626 km^2) is answered as 592.9 by every Mistral-family instruct configuration, 593 by Qwen3.5, and 580 by Gemma—stable, model-specific wrong values that survive every prompt. Three observations locate their origin in shared training corpora. First, *turnover across generations*: Qwen3.6 repairs 7 of Qwen3.5's wrong rows and breaks 11 new ones with nothing else changed; which rows a generation solves changes with its data snapshot, while the union oracle across generations reaches 89/100. Second, *unrelated training lineages solve mainstream-broken rows*: llm-jp-4-32b and Aya Expanse are far weaker overall yet each partially solves Corfu, which no mainstream-corpus model does. Third, *instruct-tuning amplifies the contaminated anchor*: under raw-completion prompting, Mistral 3.1 *Base* scores 10/12-any and 7/12-mode against its own instruct's 9/4, answers Borðoy at 20/20 where every instruct variant fails, and is the only Mistral-family configuration in which the correct Corfu value wins half the votes; the gap replicates on Llama-3.1-8B (mode accuracy 60 vs. 53/100). Base models recover specific rows while losing overall focus, so the practical system keeps the instruct model—but the closed-book ceiling is set by corpus co-occurrence statistics that instruction tuning sharpens, for better and, on contaminated values, for worse.

5 Related Work

Probing factual knowledge in LMs goes back to LAMA (Petroni et al., 2019); the LM-KBC series (LM-KBC organizers, 2023, 2024) extends it to KB construction with unconstrained cardinalities. Self-consistency (Wang et al., 2023) aggregates sampled answers by majority vote and ReWiSe (Albert-Roulhac and Zouaq, 2025) tuned it relation-wise; we extend the line to empty answer sets, numeric tolerance, surface-form merging, and cross-family arbitration. Sprague et al. (2025) find chain-of-thought helps reasoning but not knowledge recall; we refine this for recall tasks: procedures and instructions do not move retrieval, but exemplar style does, conditional on recall reliability and post-training. Ma and Hewitt (2026) report prompting moves parametric knowledge access by 1–2% and RL fine-tuning by $\sim$10%; our exemplar effect (+6% on the affected relation) sits between, and fine-tuning is prohibited here. Yuchi et al. (2026) find probe-recoverable numeric knowledge exceeds what models verbalize, one reading of our oracle–mode gaps; Kalyan et al. (2021) and Park et al. (2022) document the magnitude and unit fragilities our decimal-rule and unit-merge components target; and the grounding instructions instantiate "According to…" cues (Weller et al., 2024) for Wikidata.

6 Conclusion

We described a two-prompt, arbiter-ensembled sampling system (validation 0.641, test 0.625) and the experiments behind it. What matters, in order: how reliably the model recalls the stored value, measurable by a half-hour probe; exemplar style, the one prompt control that moves retrieval, with sign conditions we mapped; cross-family arbitration that cannot lose rows; and everything else—instructions, procedures, temperature, output routes, judges, confidence, consensus—either inert or validation-overfit. Test-side ablations across seven submissions turn the val$\rightarrow$test transfer question itself into a result: gains with mechanism-level accounts transferred, gains fitted to validation row statistics did not, and our practical advice is to demand the former before deploying anything a 100-row validation set endorses.

Limitations

The mechanism findings rest on two model families in depth (Qwen, Mistral) with single-seed generation for most prompts; seed replications exist only for the exemplar and arbitration results. The 12-row probe is stratified on one system's failures, and its ranking is validated on deployment decisions we took, not a held-out model set. Oracle and mode counts on 100-row relations carry $\pm$1–2 row noise, which bounds several negative results (e.g. arbiter increments). The base-vs-instruct comparison confounds weights with prompting interface (raw completion vs. chat); separating them needs an instruct model probed by raw completion, which chat templates make non-trivial. The contamination account is observational—consistent across three independent phenomena but not traced to specific corpus documents. Aggregation thresholds are validation-tuned; the transfer taxonomy of Section 4.6 comes from seven submissions on one task edition and its predictive use (the undeployed city prompt) is a single pre-registered case. Two deployed components (the tolerance bridge and the raw-completion capacity prompt) are validation-verified across seeds and prompt pairings but had no official test measurement at submission time. Finally, we disclose that submissions v6–v9 included a second 8B arbiter that put the ensemble at $\approx$39.8B, over the 32B cap; the violation was ours, found in a self-audit, and v10 removes it at identical validation score. Test numbers for v6–v9 should be read as at-risk under the cap rule.

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A Reproducibility

Code, configurations, builders, the step-by-step recipe (`docs/V4_REPRODUCE.md`), and per-experiment records: `https://github.com/acrosasaki-rgb/llm-knowledge-task-2026`. Model revisions, container digests, and the dataset commit are pinned; the final prediction file is a deterministic function of seven persisted candidate sets and a builder script, reproducing byte-identically on re-composition (SHA-256 verified). Candidate sets regenerate to within ±1 row under changed seeds; a 100-row $\times$ 20-candidate run costs $\sim$15 minutes on 8$\times$A100, and every comparison here is a CPU re-aggregation of frozen sets.